Feature: Formula Builder (Incentive Tier KPIs)

Field	Value
Product page	
Components	FYNXT Cloud: Global Configuration → Incentive Tier KPIs. Incentive Engine.
Design	
Phase	Phase 1 of Incentive Cloud. Next: Custom Events, Object Enrollment, PAMM Rebates.
Jira epic	
Product owner	
Needed by	
Page status	Draft
Summary	A broker admin can build KPIs from CRM object fields, as one variable or a formula, for incentive program tier calculations.

Why

The problem: KPI variables come from a fixed list (Trading Data, CRM Data), so admins can only use the fields in that list.

Who asked:

What it costs today:

What good looks like: Admins pick a CRM object and build their own KPIs from its fields, as one variable or a formula.

Who is involved

Actor	Their part in this feature
Broker admin	Creates and manages KPIs
FYNXT Cloud (CRM)	Shows the screens; supplies objects, fields and Applicable Profiles
Incentive Engine	Receives the KPI on Publish and uses it in tier calculations
Program setup	Uses published KPIs as Metrics in program conditions

User stories

US1 - Create a Variable Only KPI (Must)

As a broker admin, I want to pick one CRM object field as a KPI, so that I can use it in tier calculations.

Independent test: Publish a KPI from one field; it appears in the KPI list with KPI Type "Variable Only".

1. Given Variable Only is selected, when the admin clicks or drags a field tile, then that field becomes the selected variable.

US2 - Create a Formula Based KPI (Must)

As a broker admin, I want to combine fields with operators and functions, so that I can define KPIs that are not a single field.

Independent test: Publish a KPI with a formula; it appears in the KPI list with KPI Type "Formula Based".

1. Given Formula Based is selected, when the admin clicks a field tile, then it is inserted at the cursor as object.field.

US3 - Find and manage KPIs (Must)

As a broker admin, I want to search, filter and act on KPIs in one list, so that I can maintain them.

Independent test: Filter the list by Program Type; only matching KPIs show.

US4 – See formula errors before publishing (Must)

As a broker admin, I want basic formula errors shown while I build, so that I do not publish a KPI that cannot be calculated.

Independent test: E4, E5 and E6.

1. Given a bracket is not closed, when the admin types, then an error shows under the formula and Publish KPI is disabled.

User journey

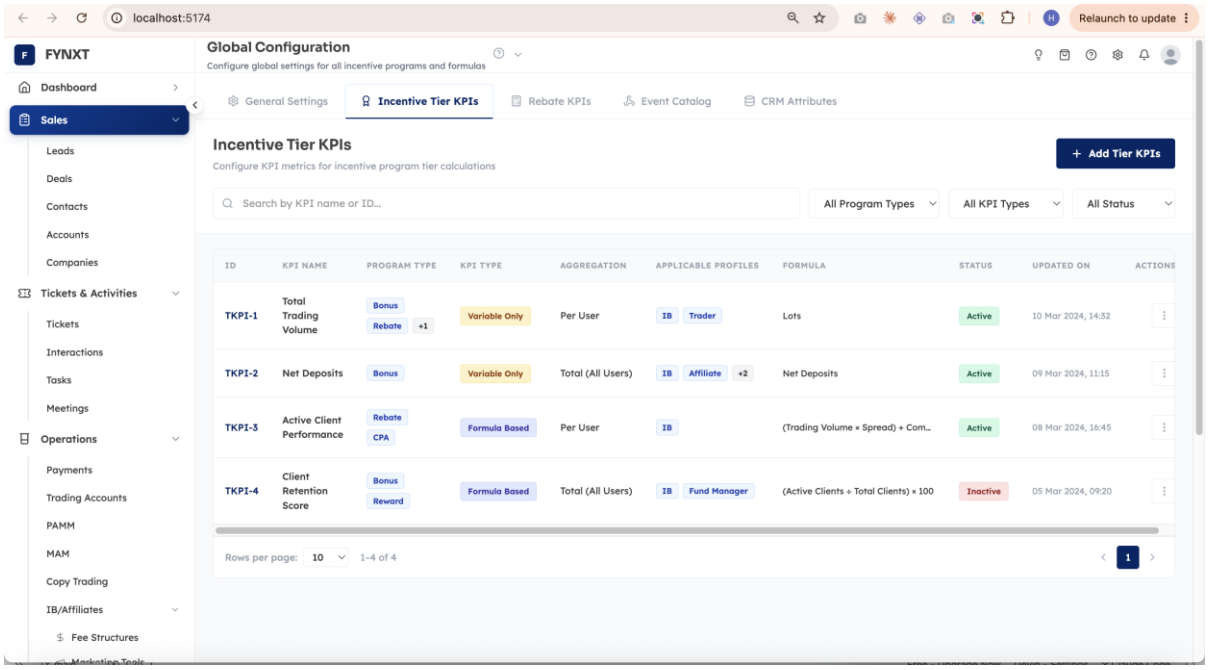
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Broker admin            FYNXT Cloud                Incentive Engine
     |--1 Add Tier KPIs-->|                             |
     |--2-4 fill form---->|                             |
     |--5 Publish KPI---->|--6 KPI definition-------->|
     |<-7 KPI in list-----|                             |
```

1. [UI] Broker admin opens Global Configuration → Incentive Tier KPIs and clicks Add Tier KPIs (S1).
2. [UI] Enters KPI Name, Program Type, Aggregation and Applicable Profiles (S2).
3. [UI] Picks a CRM object and finds fields in the Variables panel (S2).
4. [UI] Chooses KPI Type and selects a variable or builds a formula (S2). Errors show under the formula; Publish KPI stays disabled until there are none (U14, U15).
5. [UI] Clicks Publish KPI.
6. [Backend] The KPI is sent to the Incentive Engine.
7. [UI] The KPI appears in the list (S1).

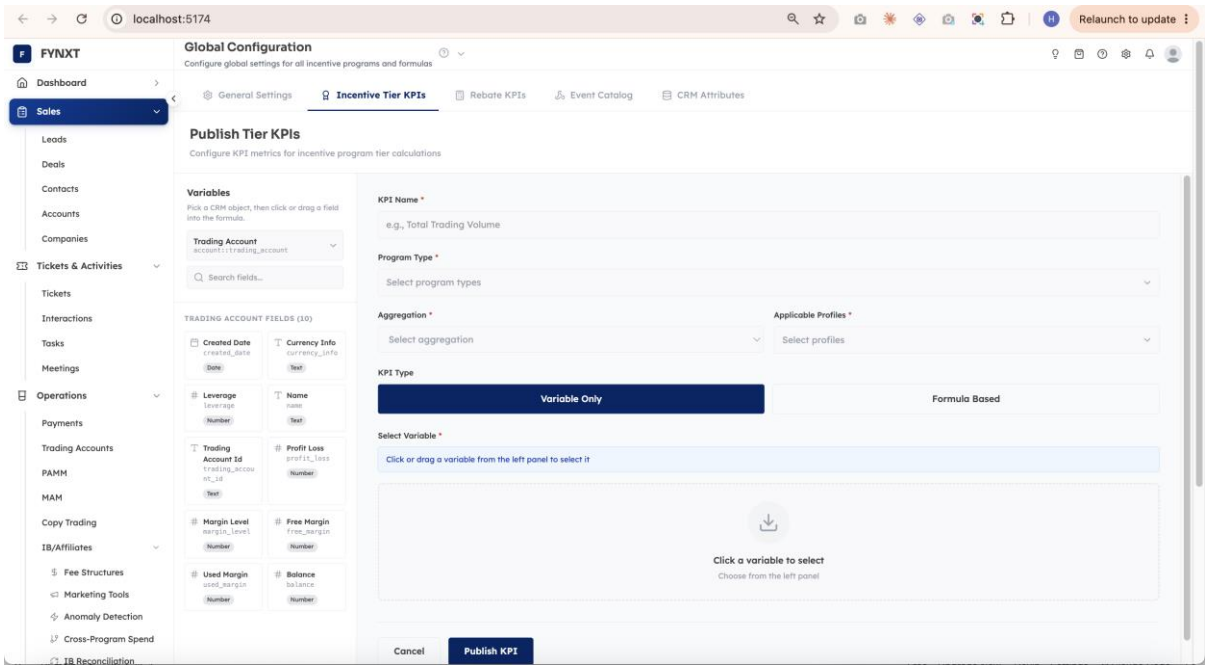
Screens

ID	Screen	Frame / prototype link	Purpose	States
S1	Incentive Tier KPIs list		See, search, filter and act on KPIs	Filled
S2	Publish Tier KPIs		Create a KPI	Variable Only, Formula Based, formula error

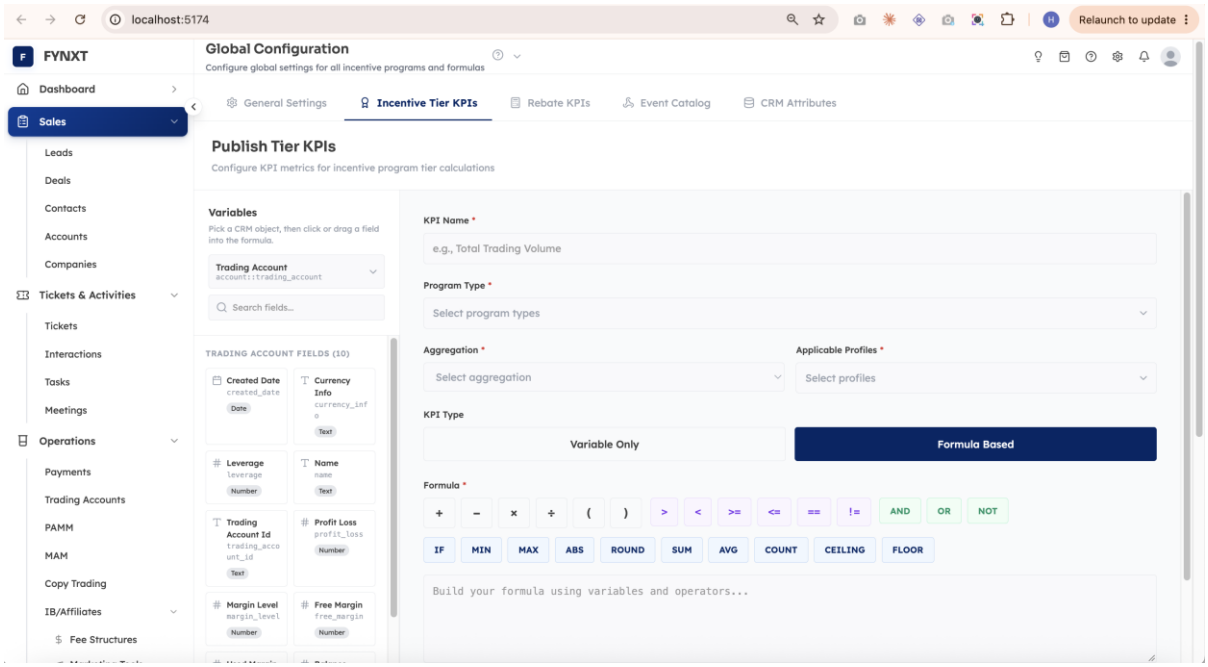
S1 · Incentive Tier KPIs list



S2 · Publish Tier KPIs, Variable Only



S2 · Publish Tier KPIs, Formula Based



Rules

Front end

U1. Required: KPI Name, Program Type, Aggregation, Applicable Profiles, KPI Type, CRM object, and Select Variable (Variable Only) or Formula (Formula Based).

U2. Program Type is multi-select. Prototype options: Bonus, Rebate, CPA, Reward, Contest.

U3. Aggregation options: Total (All Users), Per User.

U4. Applicable Profiles is multi-select: IB, Affiliate, Trader, Investor, Fund Manager.

U5. Variables panel: pick a CRM object, then search its fields. Each field shows its label, technical name and type (Number, Text, Date).

U6. Variable Only: clicking or dragging a field makes it the selected variable.

U7. Formula Based: clicking a field inserts it at the cursor as object.field, for example trading_account.balance.

U8. Formula buttons: + − × ÷ ( ) > < >= <= == != AND OR NOT, and IF, MIN, MAX, ABS, ROUND, SUM, AVG, COUNT, CEILING, FLOOR.

U9. Under the formula, a "Reads as" line shows it in plain words.

U10. List columns: ID, KPI Name, Program Type, KPI Type, Aggregation, Applicable Profiles, Formula, Status, Updated On, Actions.

U11. List: search by KPI name or ID; filter by Program Type, KPI Type and Status; rows per page.

U12. Program Type and Applicable Profiles show up to 2 chips, then "+N".

U13. Row actions: Edit, Duplicate, Activate / Deactivate, Delete.

U14. The formula must give a number.

U15. The formula is checked as it is built. A bracket not closed, an unknown field, an operator with nothing on one side, or a Text or Date field used in a calculation shows an error under the formula. Publish KPI is disabled while there is an error.

Back end

B1. CRM objects, their fields and field types come from the CRM, not a fixed list.

B2. The Applicable Profiles list comes from the CRM.

B3. Each KPI gets an ID in the format TKPI-n.

B4. On Publish, the KPI is sent to the Incentive Engine.

B5. Aggregation is applied to the result of the formula.

Examples

Setup: CRM object Trading Account (account::trading_account).

#	Rule	Input	Expected result
E1	U6	Variable Only; click the Balance tile	Balance is the selected variable
E2	U7, U9	Formula Based; click Balance, click ×, type 2	Formula: trading_account.balance * 2. Reads as: Trading Account Balance * 2
E3	U12	Program Types Bonus, Rebate, Reward	List shows Bonus, Rebate, +1
E4	U15	(trading_account.balance * 2	Error under the formula; Publish KPI disabled
E5	U14	trading_account.balance > 1000	Error under the formula; Publish KPI disabled
E6	U15	trading_account.created_date * 2	Error under the formula; Publish KPI disabled

Edge cases and errors

Case	Side	Expected result
Search finds no fields	UI	"No fields match your search."
A Text or Date field used in a calculation, e.g. created_date * currency_info	UI	Error under the formula; Publish KPI disabled (U15)
The formula gives true or false, e.g. balance > 1000	UI	Error under the formula; Publish KPI disabled (U14)
A required field is empty	UI	Publish KPI disabled (U1)

What must not change

- KPIs from the existing fixed lists (Trading Data, CRM Data) and the programs that use them keep working.
- Per User and Total (All Users) calculate as they do for today's KPIs.
- Edit, Duplicate, Activate / Deactivate and Delete behave as they do for today's KPIs.

Out of scope

- Event metrics in the Variables panel: Custom Events, Phase 2.

- Fields from more than one CRM object in one formula.
- Rebate KPIs: this page covers Incentive Tier KPIs.
- Handling CRM fields that are renamed or deleted after a KPI uses them.
- Testing a formula on sample data, and version history.

Data and interfaces

Field	Screen	Dir.	Type	Required	Allowed values and rules	Example
KPI Name	S2	In	Text	Yes		Total Trading Volume
Program Type	S2	In	List	Yes	Multi-select (U2)	Bonus, Rebate
Aggregation	S2	In	List	Yes	Total (All Users), Per User	Per User
Applicable Profiles	S2	In	List	Yes	From the CRM (U4)	IB, Trader
KPI Type	S2	In	List	Yes	Variable Only, Formula Based	Variable Only
CRM object	S2	In	List	Yes	From the CRM	account::trading_account
Variable	S2	In	Field	When Variable Only	One field	trading_account.balance
Formula	S2	In	Text	When Formula Based	U8	trading_account.balance * 2
ID	S1	Out	Text		TKPI-n	TKPI-1
Status	S1	Out	List		Active, Inactive	Active

Non-functional

Area	Requirement
Permissions	Same as access to Global Configuration today
Speed	
Audit	Same as other Global Configuration changes today

Success criteria

Measure	Target value

Integrations

System	Direction	Change	Owner agreed?
CRM	In	Objects, fields, field types, Applicable Profiles	
Incentive Engine	Out	KPI definition on Publish	

Definition of done

☐ Screens S1-S2 match the prototype, in every listed state
☐ Rules U1-U15 and B1-B5 are met
☐ Every acceptance scenario, example and edge case passes as an automated test

☐ Everything under What must not change still passes
☐ Product has signed off

Changes after Ready for build

None

Glossary

Term	Meaning
KPI	A metric used in incentive program tier calculations
Metric	How a published KPI appears in program conditions
CRM object	A record type in the CRM, e.g. Trading Account
Field	One attribute of a CRM object, e.g. balance
Variable Only	A KPI made of one field
Formula Based	A KPI made of fields, operators and functions
Aggregation	How values are combined: Total (All Users) or Per User
Applicable Profiles	The participant types a KPI applies to